

Community Nutrition and Lifestyle Medicine for Kidney Health: Awareness of the Social Determinants of Health in Renal Nutrition



COMMUNITY NUTRITION IS the practice of nutrition education and implementation among groups of persons with the goal of promoting health and wellness and preventing or managing diseases by means of nutritional and dietary interventions at community level.¹ It involves identification of unmet nutritional needs in a given community in order to design and apply nutrition and diet-related approaches to improve public health nutrition and health outcomes. Target communities can be diverse with different settings including workplaces or education centers such as schools, specific racial/ethnic groups, or persons at high risk of certain disease states such as those at risk of chronic kidney disease (CKD). Nutrition education is the core component of community nutrition efforts and includes food preparation skills. This encompasses culinary medicine approaches and food preparation methods based on the concept of food as medicine.² Practice of Community nutrition is often aligned with strategies under the lifestyle medicine.³ The latter is defined as a medicine specialty that promotes lifestyle modifications including nutritional and dietary approaches as preventive and therapeutic interventions to manage chronic conditions including obesity, metabolic syndrome, or diabetes mellitus. Cardiovascular and renal diseases are also among targets of lifestyle medicine.³ The American College of Lifestyle Medicine has announced dietary lifestyle position for prevention (?), treatment and potential reversal of disease, stating that eating a plant-based diet is recommended predominantly on a variety of minimally processed vegetables, fruits, whole grains, legumes, nuts, and seeds.⁴

Social determinants of health including equitable access to nutrition education and food security impact the field of community nutrition.⁵ Registered dietitian interested in engaging in community nutrition practices can take advantage of standards of practice under the “Public Health and Community Nutrition Dietetic Practice Group” guided by the Academy of Nutrition and Dietetics Quality

Management Committee using a toolkit that includes 4 steps consisting of assessment, diagnosis, intervention, and evaluation/monitoring in Community Nutrition.⁵ This and other toolkits can align the scope of expertise of dietitians including in medical nutrition therapy (MNT) to expand to the field of community nutrition.⁶⁻⁹

Proper nutrition and lifestyle choices play a crucial role in maintaining kidney health and preventing the onset or progression of CKD. Community nutrition initiatives and lifestyle medicine approaches have gained recognition as effective strategies to promote kidney health within populations. Practical tips for individuals and communities to adopt healthier habits can start with understanding the concept of a kidney health–friendly lifestyle and nutrition. Community nutrition education can start by providing an overview on the impact of poor lifestyle choices, such as unhealthy diets on kidney function.¹⁰ Community nutrition initiatives include discussing the significance of community-based nutrition programs in promoting kidney health with emphasis diet quality index,¹¹ management of diabetes and obesity with concurrent CKD,⁹ culinary medicine and food as medicine,⁶ precision nutrition approaches in community nutrition,¹² accurate estimates of renal function,¹³ use of plant dominant diets,¹⁴ nutrition in living kidney donors,¹⁵ role of dietitians in medical nutrition therapy to slow CKD progression,^{6,8,9} health literacy and nutrition literacy in kidney care,¹⁶ the role of social media in renal nutrition,¹⁷ and the role of functional nutrition, naturopathic nutrition, and integrative and holistic renal nutrition in kidney health.¹⁸

Highlighted initiatives can encompass community educational workshops that aim to increase access to fresh, nutritious foods, and improve nutritional knowledge within local communities at risk of CKD. To that end the importance of collaboration between healthcare professionals, community organizations, and policymakers should underscore effective approaches to implement and sustain these initiatives. A kidney–friendly diet for promoting kidney health includes moderating animal protein consumption including plant–dominant low protein diets (also known as PLADO), explain how community nutrition programs can help individuals make healthier food choices, provide resources for food preparation and meal planning, grocery shopping, and cooking kidney–friendly meals. Community involvement, support networks, and

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education programs are important in promoting kidney health and fostering behavior change, and sustaining healthy habits, highlighting the value of support groups and peer-to-peer mentoring for individuals living with kidney disease. Finally, assessing social determinants of health and promoting equitable access devoid of vestiges of structural racism¹³ is of essence to address disparities in kidney health outcomes within communities. In this issue of the Journal, Kalantar-Zadeh et al¹⁹ provide a consensus statement on assisting patients who want to transition slowly toward renal replacement therapy. The focus is on patient empowerment and utilizing dietary strategies together with pharmacotherapies to support a conservative approach, which can benefit from community nutrition initiatives. Kowal et al²⁰ performed a prospective study of patient acceptance of a mobile application providing diet information for patients on hemodialysis. Patients below 50 years of age were more likely to utilize the app than older patients and women were more accepting than men. Wakasugi et al²¹ performed an evaluation of dietary intake of fruits and vegetables in over 2000 patients in a hospital outpatient clinic and associated the outcomes with mortality. Lower fruit and vegetable intake was positively associated with lower kidney function and risk for all-cause mortality. These findings can be shared with larger groups of persons in the context of community nutrition programs.

de Paula et al²² report on the physical function of patients on hemodialysis compared to healthy controls using three types of sit-to-stand (STS) tests and handgrip strength tests. Patients on hemodialysis were slower in the STS performance than controls and the 10-repetition test was most closely associated with handgrip strength. Shah et al²³ examined whether an association exists between cystatin C or serum creatinine and appendicular lean muscle index (APMI) from whole body dual-energy absorptiometry as an assessment of sarcopenia. They utilized data from the National Health and Nutrition Examination Survey in adults and found that the kidney function markers did not improve the relationship with reduced APMI better than age, sex, and body mass index. Parker et al²⁴ performed a prospective assessment of the use of a high-protein whole food snack after hemodialysis sessions taken over a 6-month period. In patients with serum albumin <4 g/dL, high consumers of the snack experienced qualitative increases in serum albumin. Iida et al²⁵ examined the impact of changes in abdominal fat on mortality in patients on maintenance hemodialysis. They determined that visceral fat and ratios of visceral fat to subcutaneous fat were independently associated with mortality over a 60-month follow-up period. Fang and colleagues²⁶ provide a review of the literature regarding the relationship of accumulation of sodium in muscle to muscle function in patients with CKD. The review demonstrates the paucity of data on the mechanisms for increased muscle sodium

in CKD, and proposes areas for further study that may lead to improved muscle function in this population. Importance of muscle health in CKD can be more effectively communicated in nutrition community programs.

Pourghazi et al¹⁰ performed a systematic review on the literature of childhood obesity and its impact on kidney function in later years. They found a significantly positive association of childhood obesity for adult kidney disease in 75% of the examined studies. Lee et al²⁷ applied nutrition assessment using Subjective Global Assessment (SGA) criteria to examine rate of decline in glomerular filtration rate (GFR) of 236 patients with autosomal dominant polycystic kidney disease. The study demonstrated that SGA 3–6 at baseline was associated with >3 mL/min/1.73 m² decline in estimated GFR regardless of the presence of proteinuria. Kelly et al²⁸ provide a review of the status of using digital and virtual technologies in providing nutrition care of patients with kidney disease. They review methods that can be applied during each phase of nutrition therapy and critique the potential barriers and benefits of each. Digital and virtual technologies can be important components of future programs in the context of community nutrition in kidney medicine.

In conclusion, community nutrition and lifestyle medicine approaches can promote kidney health significantly. Individuals and communities should be encouraged to embrace healthier lifestyles, make informed dietary choices, and engage in regular physical activity to support kidney health. The role of community support, education, and policy changes should be emphasized in creating an environment that prioritizes Community Nutrition for kidney health for all.

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